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Improvements in the analytical methodology for the residue determination of the herbicide glyphosate in soils by liquid chromatography coupled to mass spectrometry

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ABSTRACT

The determination of glyphosate (GLY) in soils is of great interest due to the widespread use of this herbicide and the need of assessing its impact on the soil/water environment. However, its residue determination is very problematic especially in soils with high organic matter content, where strong interferences are normally observed, and because of the particular physico-chemical characteristics of this polar/ionic herbicide. In the present work, we have improved previous LC–MS/MS analytical methodology reported for GLY and its main metabolite AMPA in order to be applied to “difficult” soils, like those commonly found in South-America, where this herbicide is extensively used in large areas devoted to soya or maize, among other crops. The method is based on derivatization with FMOC followed by LC–MS/MS analysis, using triple quadrupole. After extraction with potassium hydroxide, a combination of extract dilution, adjustment to appropriate pH, and solid phase extraction (SPE) clean-up was applied to minimize the strong interferences observed. Despite the clean-up performed, the use of isotope labelled glyphosate as internal standard (ILIS) was necessary for the correction of matrix effects and to compensate for any error occurring during sample processing. The analytical methodology was satisfactorily validated in four soils from Colombia and Argentina fortified at 0.5 and 5 mg/kg. In contrast to most LC–MS/MS methods, where the acquisition of two transitions is recommended, monitoring all available transitions was required for confirmation of positive samples, as some of them were interfered by unknown soil components. This was observed not only for GLY and AMPA but also for the ILIS. Analysis by QTOF MS was useful to confirm the presence of interferent compounds that shared the same nominal mass of analytes as well as some of their main product ions. Therefore, the selection of specific transitions was crucial to avoid interferences. The methodology developed was applied to the analysis of 26 soils from different areas of Colombia and Argentina, and the method robustness was demonstrated by analysis of quality control samples along 4 months.

1. Introduction

Glyphosate [N-(phosphonomethyl)glycine] (GLY) is a broad spectrum, non-selective post-emergence herbicide widely used around the world for weed and vegetation control. When this herbicide reaches the soil, it is strongly sorbed to soil components, such as clay, iron oxides, or humic substances [1–3]. Its degradation in the soil-environment mainly occurs under biological conditions yielding aminomethylphosphonic acid (AMPA) as the major metabolite [4–6]. Although GLY would not be expected to be transported to ground and surface water, due to its strong sorption onto the soil and its microbiological degradation, both GLY and AMPA have been found in natural waters [7–9]. The urban contribution of GLY and AMPA to wastewaters has also been reported [10], as well as the presence of substantial amounts in sewage sludge from urban areas [11].

Although glyphosate has lower ecotoxicological potential than many other herbicides, a thorough assessment of its environmental occurrence is necessary given to its worldwide application. This is of special relevance in countries like Argentina, where huge areas are devoted to transgenic varieties of glyphosate tolerant soybean [12], or like Colombia, where this herbicide is also widely used in agriculture. The environmental fate and behaviour of glyphosate is subjected to controversial. One of the main problems comes from its difficult determination in soil and water samples. Its amphoteric and highly polar character [13] makes its residue determination an analytical challenge, particularly in soil samples with high organic matter content, due to their higher complexity and likely presence of interferent compounds.

Most methods reported for GLY and AMPA are troublesome, with tedious sample preparation, and a derivatization step is

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normally required for its appropriate chromatographic determination. The wide majority of early methods made use of a derivatization step with fluorenylmethyl chloroformate (FMOC) to produce a fluorescent derivative. The use of coupled-column liquid chromatography (LC–LC) with fluorescence detection allowed in the 90s to reach the sensitivity and selectivity required for GLY residue determination in water [14,15], soil [16] and vegetable products [17]. Although this approach was an important advance, the irruption and instrumental developments of LC coupled to tandem mass spectrometry (LC–MS/MS) has allowed renewing and improving the previous methodology based on the use of conventional detection systems. However, even using LC–MS/MS, the derivatization of GLY and AMPA is normally required to reach sufficient retention into the reversed-phase chromatographic columns commonly used and/or for analyte pre-concentration in most chromatographic sorbents. In the last years several papers have been published on GLY and AMPA determination in environmental samples, mainly soil and water, making use of LC–MS, or LC–MS/MS, previous derivatization with FMOC [8,9,18–20]. This technique has also been used for their determination in sewage sludge after FMOC derivatization on strong anion-exchange resin as solid support [21]. Due to the high complexity of the samples, purification on SAX resin was firstly required followed by a pre-concentration of the derivatized analytes on SPE Oasis HLB cartridges. Although a few methods have recently reported the direct determination of GLY (i.e. without derivatization) in environmental, food and biological matrices, using HILIC columns, designed for very polar analytes [22,23], anion-exchange [24] or even reversed-phase columns [25], however there is not yet a methodology widely accepted using this type of columns, and more research is required to demonstrate the robustness when applied to complex-matrix samples.

From our first works on GLY residue analysis based on LC–LC/FD [14–17], we moved to LC–MS/MS, which was successfully applied to water and soil samples from Spain [8,9]. However, some difficulties were found when we analyzed soils collected from South America, most of them characterized by high organic matter content. Severe matrix effects and poor reproducible data were observed, which encouraged us to update the analytical methodology for this type of samples. In the present work, a detailed study has been made on the analytical determination of GLY and AMPA residues in "difficult" soil matrices, making use of LC–MS/MS with triple quadrupole. After soil extraction with KOH, the need of performing an efficient clean-up (either before or after derivatisation) has been evaluated. Several possibilities were tested in order to find the best option in terms of sensitivity, matrix effects and robustness. Special attention has been paid to the presence of matrix components that affect not only the ionization of analytes but also interfere the MS/MS transitions monitored. To this aim, LC–QTOF MS has been a valuable tool thanks to the accurate-mass full-spectrum acquisition data provided by this technique.

2. Experimental

2.1. Reagents and chemicals

Glyphosate (98%) and AMPA (99%) reference standards were purchased from Dr. Ehrenstorfer (Augsburg, Germany) and Sigma (St. Louis, MO, USA), respectively. Isotope-labelled glyphosate (1,2-^{13}C, ^{15}N), used as internal standard (IS), was purchased from Dr. Ehrenstorfer. Analytical reagent-grade disodium tetraborate decahydrate was obtained from Scharlab (Barcelona, Spain) and 9-fluorenmethylcholoroformate (FMOC-Cl) was purchased from Sigma. Reagent-grade hydrochloric acid, formic acid, potassium hydroxide (KOH), acetic acid (HAc) and ammonium acetate (NH_4Ac) as well as LC-grade acetonitrile were purchased from Scharlab. HPLC-grade water was obtained by water passed through a MilliQ water purification system (Millipore Ltd., Bedford, MA, USA).

Standard stock solutions were prepared dissolving approximately 50 mg powder, accurately weighted, in 100 mL of water obtaining a final concentration of approximately 500 mg/L. A 50-mg/L composite standard was prepared in water by mixing and diluting the individual standard stock solutions. Standard mixed working solutions for LC–MS/MS analysis and for samples fortification were prepared by dilution of the 50-mg/L composite standard with water.

The isotope-labelled glyphosate used as internal standard (ILIS) was purchased as 1.1 mL of 100-μg/mL stock solution in water. A 11-μg/mL ILIS solution was prepared by dissolving 1.1 mL of the stock solution in 10 mL of water. ILIS working solutions were prepared by diluting the intermediate standard solution with water.

Solutions of 5% (w/v) borate buffer (pH approximately 9) in HPLC-grade water and 12,000 mg/L of FMOC-Cl in acetonitrile were used for the derivatization step. OASIS HLB cartridges (60 mg) as well as OASIS MAX cartridges (60 mg) were purchased from Waters (Mildford, MA, USA). 0.45 μm nylon filters were purchased from Scharlab.

2.2. Instrumentation

Ultra-performance liquid chromatography (UPLC^{TM}) system Acquity (Waters, Milford, MA, USA) was interfaced to a triple quadrupole mass spectrometer (TQD, Waters Micromass, Manchester, UK) using an orthogonal Z-spray-electrospray interface. The LC separation was performed using a Discovery analytical column C_{18}, 5 μm particle size 50 mm × 2.0 mm i.d. (Supelco, Bellefonte, PA, USA), at a flow rate of 300 μL/min. Mobile phase was a time-programmed gradient using A (H_2O 5 mM acetic acid/ammonium acetate, pH 4.8) and B (acetonitrile). The percentage of organic modifier (B) was changed linearly as follows: 0 min, 10%; 2.5 min, 10%; 2.6 min, 90%; 4.6 min, 90%; 4.7 min, 10%; and 10 min, 10%. The injection volume was 20 μL. Drying as well as nebulizing gas was nitrogen, obtained from a nitrogen generator. The gas flow was set to 1200 L/h. For operation in MS/MS mode, collision gas was Argon 99.995% (Praxair, Madrid, Spain) with a pressure of approximately 4.10^{-3} mbar in the collision cell. Experiments were performed in positive ionization. A capillary voltage of 3.5 kV was applied. The desolvation gas temperature was set to 350 °C and the source temperature to 120 °C. Temperature column was set to 40 °C. Dwell times of 0.20 s/scan were chosen. MassLynx v 4.1 (Waters, Manchester, UK) software was used to process the quantitative data obtained from calibration standards and from samples.

Waters Acquity UPLC (Waters) was interfaced to a hybrid quadrupole-orthogonal acceleration-TOF mass spectrometer (Q-oaTOF Premier, Waters Micromass), using an orthogonal Z-spray-ESI interface. TOF-MS resolution was approximately 10,000 at full width half maximum (FWHM) at *m*/*z* 556.2771. LC conditions were the same indicated above. Capillary voltage of 3.5 kV and cone voltage of 25 V were used. For MS/MS experiments, collision energy of 10 eV was selected. For automated accurate mass measurement, the lock-spray probe was used, using as lockmass leucine enkephalin (2 μg/mL) in acetonitrile:water (50:50) at 0.1% HCOOH.

2.3. Recommended procedure

The procedure was based on the previously reported [8], although including a clean-up step consisting on a first dilution of the sample extract, an adjustment of the pH to 9 and an SPE step using Oasis HLB cartridges. The procedure applied was as follows: 2.0 g soil sample (previously dried at room temperatures

Table 1
Optimized 1 MS/MS parameters for GLY, AMPA and ILIS.

Compound	Precursor ion	Product ion	Q/q ratio**	Cone voltage (V)	Collision energy (eV)
GLY-FMOC	392.3	Q 88.0	–	20	25
		q_1 170.0	2.66		15
		q_2 179.3*	2.25		20
		q_3 214.1	1.87		10
AMPA-FMOC	334.2	Q 179.2*	–	20	25
		q_1 112.0	2.59		15
ILIS-FMOC	395.3	Q 91.0	–	20	25
		q_1 173.0*	–		15
		q_2 179.3*	–		20
		q_3 217.1*	–		10

* Transitions interfered in several of the soils tested.
** Average value for standards used in the calibration curves.

and homogenized) was weighted into a 50-mL centrifuge tube. The sample was extracted with 10 mL 0.6 M KOH on a mechanical shaker for 30 min, and then centrifuged at 3500 rpm for 10 min. Then, 1 mL of the supernatant was diluted with 1 mL HPLC-grade water into a glass tube, and 200 μL of the isotope labelled internal standard (ILIS) (1.10 mg/L) was added. The soil extract was adjusted to pH 9 by adding HCl 6 M and/or 0.6 M, and it was loaded onto a OASIS HLB cartridge (60 mg), previously conditioned by passing 2 mL methanol and 2 mL water at pH 9. The non-retained sample extract was collected and derivatized with 120 μL borate buffer and 120 μL of FMOC-Cl reagent. The tube was swirled and left overnight at room temperature (between 12 and 15 h). After that, the derivatized extracts were filtered through a 0.45 μm nylon filter and acidified with HCl (c) to pH 1.5 and let stand for 1 h. Then, it was centrifuged, and 20 μL of the final extract was injected into the LC–ESI-MS/MS system under the chromatographic and MS conditions indicated in instrumentation and Table 1, respectively. A scheme of the procedure applied is depicted in Fig. 1 (procedure B).

2.4. Validation study

Validation of the method was based on European Union SANCO guidelines [26,27]. Precision (repeatability, in terms of % RSD) and accuracy (percentage recoveries) were estimated by recovery experiments in four selected soils (two from Argentina and two from Colombia), at two fortification levels each (0.5 and 5 mg/kg), and analyzed in quintuplicate. Non-spiked soils were also analyzed in duplicate to obtain the concentrations of GLY and AMPA in the "blank" samples. Recoveries between 70%-120%, with RSD lower than 20%, were considered as satisfactory.

Linearity of the method was evaluated analysing five standard solutions, in duplicate, in the range of 5–2500 μg/L. Linearity was considered satisfactory when correlation coefficient was higher than 0.99 and residuals lower than ±20%.

The specificity of the method was evaluated by analysing a procedure blank, a processed "blank" soil, and a processed "blank" soil spiked at the lowest concentration level tested. Due to the presence of both GLY and AMPA in the soils used in validation experiments, the most common approach, e.g. absence of chromatographic peaks at the MS/MS transitions monitored, could not be applied. Instead, a detailed study on several soils by monitoring all available transitions was made together with QTOF MS analysis to evaluate the possible presence of matrix interferents making use of the accurate-mass full-acquisition data generated.

The limit of quantification (LOQ) was estimated for a signal-to-noise ratio of 10 from the chromatograms of samples spiked at the lowest concentration validated (i.e. 0.5 mg/kg), making use of the quantification transition (Q). The limit of detection (LOD) was estimated similarly to the LOQ but for a signal-to-noise ratio of 3 from the chromatograms of samples spiked at the lowest concentration validated.

Confirmation of the identity of the compound in samples was carried out by acquisition of four (GLY) and two (AMPA) MS/MS available transitions, and the compliance of at least one *Q*/*q* ratio when compared with reference standards (*Q*, quantification transition; *q*, confirmation transition). Maximum deviations accepted in *Q*/*q* ratios were based on guideline SANCO/10684/2009 [26] as well as on European Decision 2002/657 [28] and varied between 10% and 50% depending on the *Q*/*q* ratio value. The agreement in retention time between standards and samples was also required, with maximum deviation of 2.5%.

3. Results and discussion

Experiments for method development, optimization and validation were carried out on several soils collected in Colombia and Argentina from agricultural areas where glyphosate is widely used. Around 1 kg soil sample was collected and dried at room temperature during at least 4 days. Then, it was homogenized and passed trough a 1-mm sieve. An aliquot of around 25 gwas introduced in 50-mL polyethylene tubes and transported to Spain, within 48 h. Five out of the six soils had acidic pH, around 4.2–4.5, and organic matter was above 3% in three of them. Physico-chemical properties and the classification of these soils are shown in Supplementary Information (Table S1, SI).

Analysis of these soils using the methodology previously developed for GLY [8] showed strong matrix effect and, in general, unsatisfactory correction using isotope-labelled glyphosate as ILIS. Poor reproducibility was also observed together with notable interferences. Therefore, in this paper we focused our work on the clean-up of the soil extracts in order to minimize matrix interferents. A detailed study of interferent compounds that might affect analytes quantification and confirmation was also made. Alkaline extraction with KOH was selected, as both NaOH and KOH have been widely applied for GLY extraction in soils, sediments and sewage sludge [8,11,16,21,29,30]. As regards the derivatization reaction, it was applied as in our previous work [8]. Although some authors reported low reaction kinetic and yield under low acetonitrile content [31], we did not observed significant problems when performing the derivatization under the conditions tested when performing the derivatization overnight (final ACN content around 5%). A filtration was required after derivatization (see recommended procedure and Fig. 1), surely as a consequence of the low solubility of FMOC in water, but it seemed not to much affect the efficiency or reproducibility of the derivatization.

3.1. LC–MS/MS conditions: study of matrix interferences

Initial conditions regarding derivatization, chromatographic and MS/MS conditions were taken from our earlier works [8].

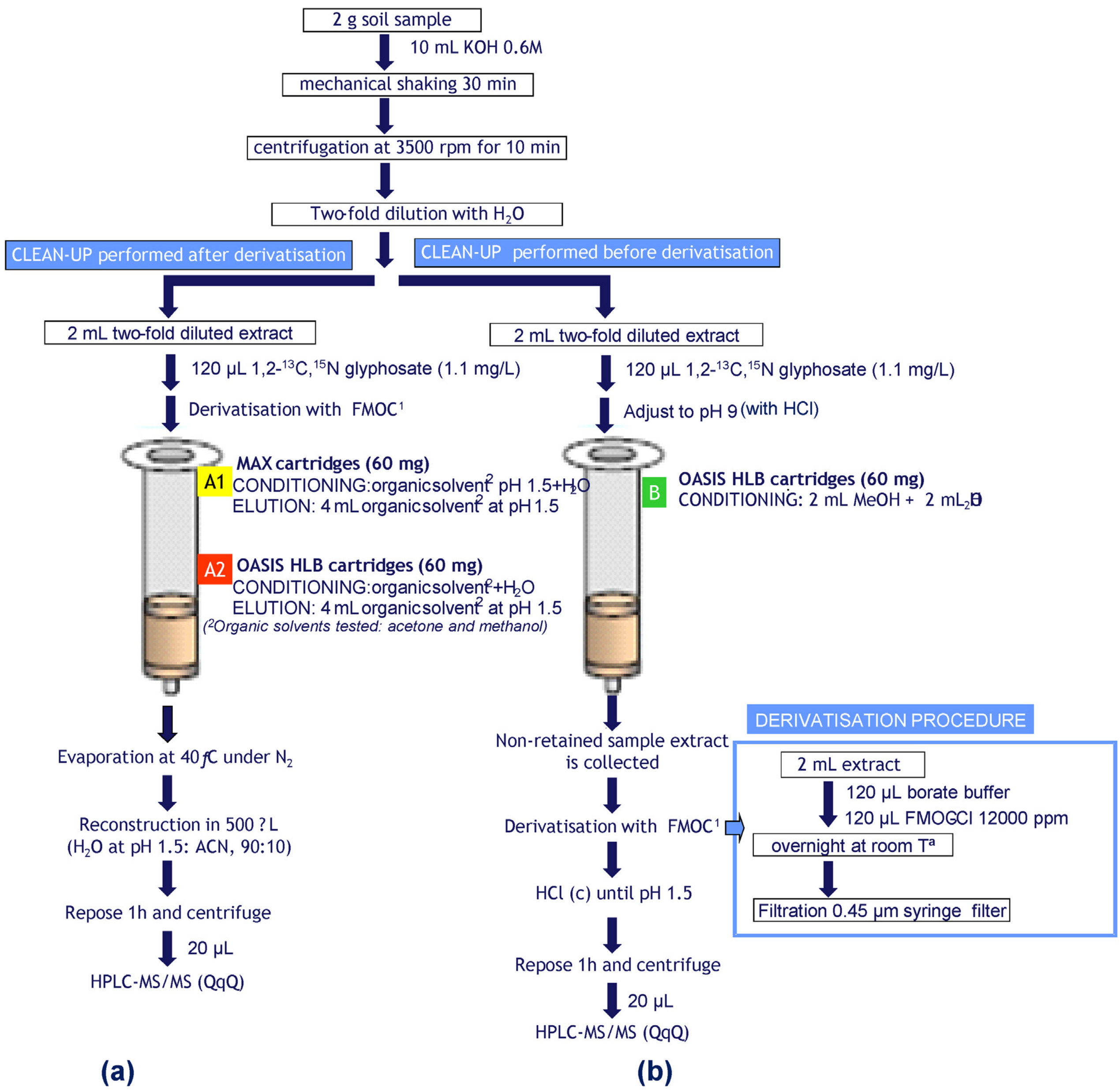

Fig. 1. Clean-up procedures assayed in this work: (a) Clean-up performed after FMOC derivatisation; (b) clean-up performed before FMOC derivatisation (recommended procedure).

Working under positive mode, two transitions for AMPA-FMOC and four transitions for GLY-FMOC were selected (Table 1). Regarding ILIS-FMOC, four transitions were also available, similarly to GLY-FMOC. Special attention was paid to the selectivity of these transitions, as our preliminary data in selected soils showed that some transitions were shared by unknown matrix components. This situation would make unfeasible the confirmation of positives, as Q/q ratios would not fit with reference standards, and might also affect the quantification (e.g. an interference in the Q transition would lead errors by excess). It is worth to notice that after derivatization with FMOC, application of a highly selective technique as LC–MS/MS, and even after applying a clean-up step (as demonstrated subsequently in this work), one of the GLY transitions (392 > 179, named as q_2) and one of the AMPA transitions (334 > 179, named as Q) were frequently interfered, especially in the soils from Colombia. In addition, up to three transitions selected for the ILIS were also shared by soil matrix components (395 > 173 (q_1), 395 > 179 (q_2), 395 > 217 (q_3)). The latter would only affect the confirmation of ILIS in the samples, which seems unnecessary as the objective of ILIS is to facilitate matrix effect correction, and to this aim only the quantification transition is required (to obtain the relative areas analyte/ILIS). However, this situation illustrates that occurrence of interferences in the LC–MS/MS determination of glyphosate in soils is more likely than expected, and the analyst must be aware of this possibility to avoid unexpected problems.

At this point, UHPLC-QTOF experiments were performed to obtain more information on the interferences observed. Next, the ILIS interference is discussed as illustrative example. The accurate mass of the protonated molecule of this unknown was found to be m/z 395.1624, which differed 69.8 mDa from the exact mass of the ILIS-FMOC (see Fig. 2). The MS/MS spectrum of this compound presented product ions at m/z 217.0860, 179.0874, 173.0937, 116.0735 and 70.0659 (Fig. 2, bottom). Product ions at m/z 217, 179 and 173 (in nominal mass) also appeared in the MS/MS spectra of ILIS-FMOC (Fig. 2, top), which explain the interference observed. Moreover, the product ion m/z 179 had the same exact mass than that of

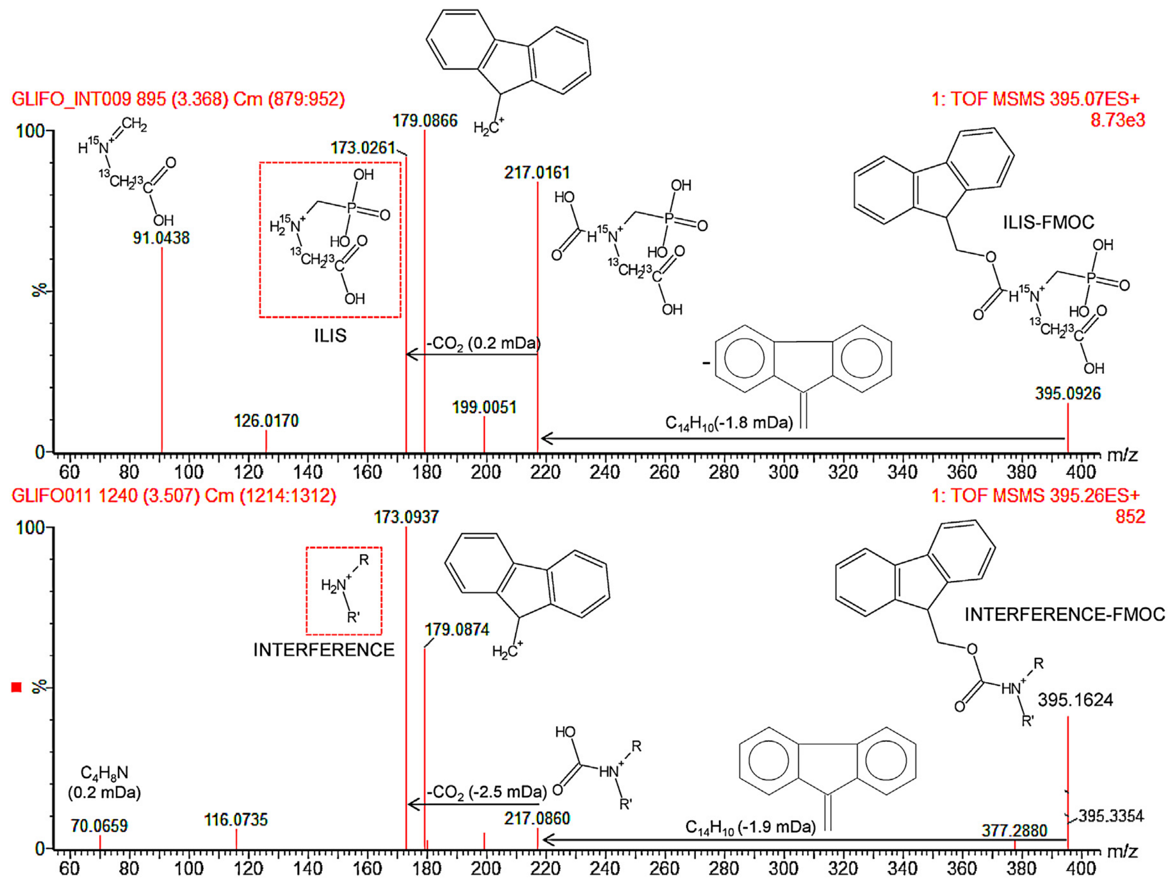

Fig. 2. LC–QTOF MS/MS spectra of ILIS-FMOC (top) and of soil interference (bottom).

ILIS-FMOC, and corresponded to $C_{14}H_{11}$ (m/z 179.0861) generated from FMOC reagent. A CO_2 loss (217→173) was also observed in both spectra. Comparing both MS/MS spectra, it seems that the unknown compound was a primary or secondary amine, which was also derivatised with FMOC. Similarly to the ILIS-FMOC spectrum, where the ion at m/z 173 corresponds to the protonated molecule of ILIS, the ion at m/z 173.0937 in the interference-FMOC spectrum, should correspond to the protonated molecule of the interference.

An attempt to elucidate this unknown was made following the strategy previously reported [32]. Possible elemental compositions for m/z 173.0937, with a maximum deviation of 3 mDa from the measured mass, were calculated using the elemental composition program within the MassLynx software. A minimum of 1 N, 4 C and 8 H atoms were considered taking into account the general structure depicted in Fig. 2 for this ion as well as the product ion at m/z 70.0659 (C_4H_8N as the only possibility). Three plausible elemental compositions were finally suggested for m/z 173.0937: $C_4H_{14}N_2O_4F$, $C_5H_{10}N_6F$ and $C_6H_{14}N_4P$ (expressed as protonated molecule). After searching in Chemical databases (Reaxys and ChemSpider), no hits were found for $C_4H_{13}N_2O_4F$ and $C_5H_9N_6F$. The formula $C_6H_{13}N_4P$ resulted in one, but none chemical structure supported by the fragment ions observed could be finally proposed (more details in Supplementary Information).

It is noteworthy that those transitions including the product ion m/z 179 were interfered in several of the soils analyzed for GLY and AMPA, and also for ILIS (Table 1). Interestingly, this product ion comes from FMOC (see Fig. 2) and consequently it has no relationship with the chemical structure of the analytes. Therefore, the specificity of the transitions needs to be taken into account, and one should avoid those transitions resulting from non-selective losses, e.g. water, CO_2, HCl, or in this case the derivatized reagent FMOC [33]. The interference occasionally observed for GLY-FMOC (transition named as q_2) was not much problem for confirmation of GLY as there were another two confirmation transitions available. On the contrary, the confirmation of AMPA was problematic in several soils, as one out of two available transitions was frequently interfered (334 > 179), and no more options existed for confirmation based on Q/q ratio tolerances, as discussed in the following sections.

Under these circumstances, the acquisition of all available transitions (instead of only two, which is the most usual approach in the literature), seems the best option to have more possibilities to find non-interfered transitions for analysis of GLY and AMPA in soils.

3.2. Clean-up study

Several approaches were tested on soil alkaline extracts in order to minimize the strong matrix effects observed: dilution of the extract with water, modification of pH, and application of SPE clean-up step. As shown in this section, a combination of the three factors was found the most appropriate to this aim. The clean-up procedures tested in this work are schematically shown in Fig. 1.

Procedure A consisted on performing the SPE clean-up after derivatization with FMOC. The objective was to retain the FMOC-derivative in the cartridge, while it was expected that interfering matrix components would pass the cartridge as unretained compounds, or would not be eluted with the eluting solvent used for analytes. Two different cartridges were tested: Oasis MAX, designed to achieve higher selectivity and sensitivity for extracting acidic compounds with anion-exchange groups, and Oasis HLB, an universal polymeric reversed-phase sorbent developed for the extraction of a wide range of acidic, basic, and neutral compounds.

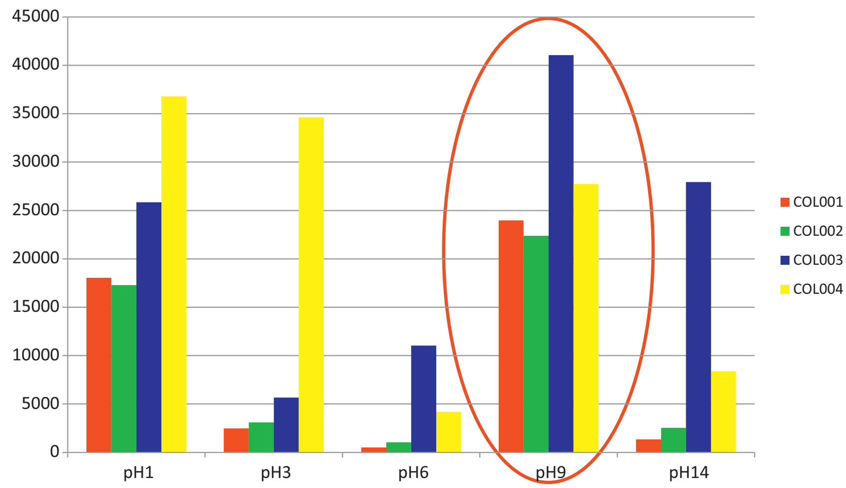

Fig. 3. Effect of the pH of the soil extract on the absolute GLY response for four soils extracts subjected to clean-up with Oasis HLB.

Using Oasis MAX (procedure A1), the results were not satisfactory. A possible explanation could be that analytes-FMOC derivatives remained in the cartridge, as low responses were obtained in the eluates. Following procedure A2 (Oasis HLB), rather satisfactory results were obtained in terms of analyte recovery (mean recoveries of 79% and 80% for GLY and AMPA, respectively). However, the robustness was poor, surely due to the sample handling and the final evaporation process for solvent exchange. The retention of the GLY-FMOC derivative in Oasis HLB and its subsequent elution with acetone or methanol seemed efficient, but this option was discarded because of the extensive sample manipulation required.

Procedure B consisted on performing the SPE clean-up before derivatization with FMOC. In this case, the analytes should pass the cartridge when loading the soil extract without being retained as a consequence of their high polarity/ionic character, while some matrix components are expected to be retained into the cartridge. Oasis HLB was only tested using this approach. A soil extract was spiked at 250 μg/L and the pH was adjusted to 1, 3, 6 or 9 before loading the extract onto the SPE Oasis HLB cartridge. The spiked alkaline extract was also loaded directly without pH adjustment (pH approx 14). Then, the extract passing through the cartridge was collected and derivatized with FMOC, as shown in Fig. 1.

The combination of adjusting pH and applying a subsequent SPE clean-up had noticeable effect at pH 1 and pH 9, leading to a notable increase in the response as a consequence of less matrix effects. Thus, an increasing factor of 15–20 in the signal was observed for the soils tested. As shown in Fig. 3, the best results in terms of increasing MS signal were obtained at pH 9 for three out of four soils tested. One of the soils (COL003) presented better response at pH 1, although the difference with respect to pH 9 was not much relevant. Therefore, adjusting the soil extract to pH 9 before SPE clean-up was selected for subsequent experiments.

Despite the clean-up applied, the use of ILIS was necessary for matrix effect correction, as recoveries around 25% were obtained when quantification was made with standards in solvent without ILIS. The ILIS was added to the soil extract before SPE to also correct for potential analyte losses that might occur during this step. After application of clean-up to soil extracts at pH 9, recoveries were rather satisfactory using ILIS correction, however a bad peak shape was observed for most of the samples. At this point, the dilution of soil extracts with HPLC-grade water was assayed as a fast and simple way to minimize matrix content. The four soil samples tested were fortified at 4.5 mg/kg and their extracts diluted 2, 5 and 10-times before derivatization. According to our results, the three dilution factors tested led to a more accurate quantification, but the dilution alone was insufficient for satisfactory data being necessary in all cases to use ILIS for quantification. Finally, a 2-fold dilution was selected as a compromise between recovery, sensitivity and peak shape.

Apart from the four soils used as a model, matrix effects were tested for many other soils by comparison of the ILIS responses in solvent and in soil extracts after the clean-up described (see Table S2, SI). With a few exceptions, strong signal suppression (higher than 70%), was still observed for most of the soils, highlighting the need of correction with ILIS. This fact is illustrative of the great difficulties of this type of sample matrices.

3.3. Method validation

The method was validated in four soils, two from Colombia and two from Argentina, following the recommended procedure (Fig. 1, procedure B).

Standards calibration showed satisfactory linearity in the studied range (5–2500 μg/L) with correlation coefficients ≥0.99 and residuals lower than 20%. This range was equivalent to 0.05–25 mg/kg in the sample.

Accuracy and precision were estimated by means of recovery experiments, by spiking the soil samples at 0.5 and 5 mg/kg, in quintuplicate. The results obtained were satisfactory for GLY in all soils, with recoveries between 92% and 107% at the 5 mg/kg level, and between 79% and 117% at 0.5 mg/kg. RSDs were below 15% in all cases (Table 2). For AMPA, recoveries were mostly between 65% and 89%, although the Colombian soils showed slightly lower values around 50%. The reason of the lower AMPA recoveries might be found on the worse peak shape and a possible poorer correction using the glyphosate ILIS.

It is interesting to remark that one of the soils used in the validation experiments (COL001) presented GLY and AMPA concentrations around 1.8 mg/kg. Therefore, the validation could not be satisfactorily made at the 0.5 mg/kg level.

LOQs and LODs were estimated from the SRM chromatograms of samples spiked at the lowest level tested. LOQs were found to be 0.05 mg/kg for GLY and 0.03 mg/kg for AMPA, while LOD were 0.02 mg/kg and 0.01 mg/kg, respectively.

3.4. Analysis of soil samples

The validated method was applied to the analysis of 26 samples from different areas of Argentina (11 soils) and Colombia (15 soils). To ensure the quality of the analysis when processing real-world samples, "blank" soil samples fortified at 0.5 and 5 mg/kg

Table 2
Method validation in different soils. Recommended 5 procedure B ($n = 5$).

Soil	Non-spiked soil		5 mg/Kg				0.5 mg/Kg			
	GLY	AMPA	GLY		AMPA		GLY		AMPA	
	mg/Kg	mg/Kg	Rec(%)	RSD(%)	Rec(%)	RSD(%)	Rec(%)	RSD(%)	Rec(%)	RSD(%)
ARG001	0.11	0.80	102	6	84	10	117	7	82	13
ARG002	0.30	0.78	92	6	89	17	79	11	75	19
COL001	1.86	1.80	102	15	53	22	–	–	–	–
COL002	0.19	0.29	107	5	48	8	82	6	65	18

-: not calculated due to the high analyte concentration in the “blank” non-spiked sample.

were used as quality controls (QC) and were distributed along the batch of samples every 4–5 injections. Taking into account the difficulties of this residue determination, the high variability of the physico-chemical properties of the soils, and the strong matrix effects that affect this type of analysis, performing a validation in a given soil seems insufficient to demonstrate the method robustness and applicability. To this aim, the use of QCs is surely the best way to give a realistic information on the method performance. In this work, six soils were used as QCs along 4 months of sample analysis. Accepting QCs recoveries between 60% and 140% [26], the results were satisfactory for GLY and AMPA at the 5 mg/kg level (81–91% GLY, and 66–119% AMPA) in the six soils tested. At the 0.5 ppm spiked level, QCs recoveries were between 52% and 83% (GLY) and between 39% and 120% (AMPA), with the lowest values being obtained for the two Colombian soils (52% and 56% GLY; 39–41% AMPA). On the contrary, at the low level of fortification the soils form Argentina seemed to be less problematic, with satisfactory recoveries, between 69% and 83% (GLY) and between 68% and 120% (AMPA). This fact support our previous findings on that Colombian soils were particularly problematic for AMPA, as illustrated by the non-compliance of the Q/q ratio for some of them. In addition, it must be taken into account that calculation of QCs recoveries at the 0.5 mg/kg is subjected to higher errors as the soils used for preparing quality controls contained GLY and AMPA. Therefore, for calculation of recoveries the amount present in the blank samples must be subtracted from the amount calculated after analysis of fortified samples.

Table 3 shows a summary of the results obtained. Both, GLY and AMPA, were found in nearly all the samples analyzed, which is in agreement with the wide use of this herbicide in both countries. Normally, the residue levels were higher in the soils from Colombia, reaching values up to 3.8 mg/kg glyphosate. This herbicide was detected in 13 out of 15 Colombian samples and the concentrations were above 1 mg/kg in 9 of these samples. As regards AMPA, it was also detected in 13 out of 15 samples, and the levels were normally lower than for GLY, reaching a maximum concentration of 2.1 mg/kg. 3 out of 15 samples contained AMPA concentrations above 1 mg/kg. In relation to the soils from Argentina, both compounds were detected in 10 out of 11 samples. GLY and AMPA concentrations in these soils were all below 1 mg/kg, with only two exceptions (AMPA concentrations in soils ARG003 and ARG004 were 1.0 and 1.2 mg/kg, respectively).

Table 3
Quantification and confirmation of GLY and AMPA in soil samples.

Soil	Glyphosate			AMPA		
	Conc (mg/kg)	Q/q_1 ratio	Q/q_3 ratio	Conc (mg/kg)	Q/q ratio (tolerance, 25%)	Q/q ratio (tolerance, 50%)
Colombia						
COL001	1.86	×	√	1.80	√	√
COL002	0.19	×	√	0.29	√	√
COL003	2.70	√		0.56	√	√
COL004	n.d.	–		n.d	–	–
COL005	0.84	√		0.55 (0.32)[a]	×	×
COL006	1.74	√		1.14	√	√
COL007	1.46	√		0.49	√	√
COL008	1.05	√		0.42 (0.29)	×	√
COL009	1.23	√		0.89 (0.67)	×	√
COL010	n.d.	–		n.d	–	–
COL011	0.78	√		0.67	√	√
COL012	1.72	√		0.81(0.83)	×	√
COL013	0.68	√		0.28(0.33)	×	√
COL014	1.34	√		0.91(0.87)	×	√
COL015	3.82	√		2.07(1.93)	×	√
Argentine						
ARG001	0.11	×	√	0.80	√	√
ARG002	0.30	√		0.78	√	√
ARG003	0.17	×	√	1.04	√	√
ARG004	0.28	×	√	1.19	√	√
ARG005	0.10	×	√	0.41	√	√
ARG006	0.41	√		0.17(0.10)	×	√
ARG007	0.49	×	√	0.30	√	√
ARG008	0.41	×	√	0.18(0.11)	×	√
ARG009	0.49	√		0.26	√	√
ARG010	0.67	√		0.69	√	√
ARG011	n.d.	–		n.d.	–	–

[a] Concentration calculated using Q and q (in brackets) transitions for those soils where AMPA was out of Q/q ratio tolerance.
(√) Ion ratio deviation within the admitted maximum tolerances.
(×) Ion ratio deviation out of admitted maximum tolerances.

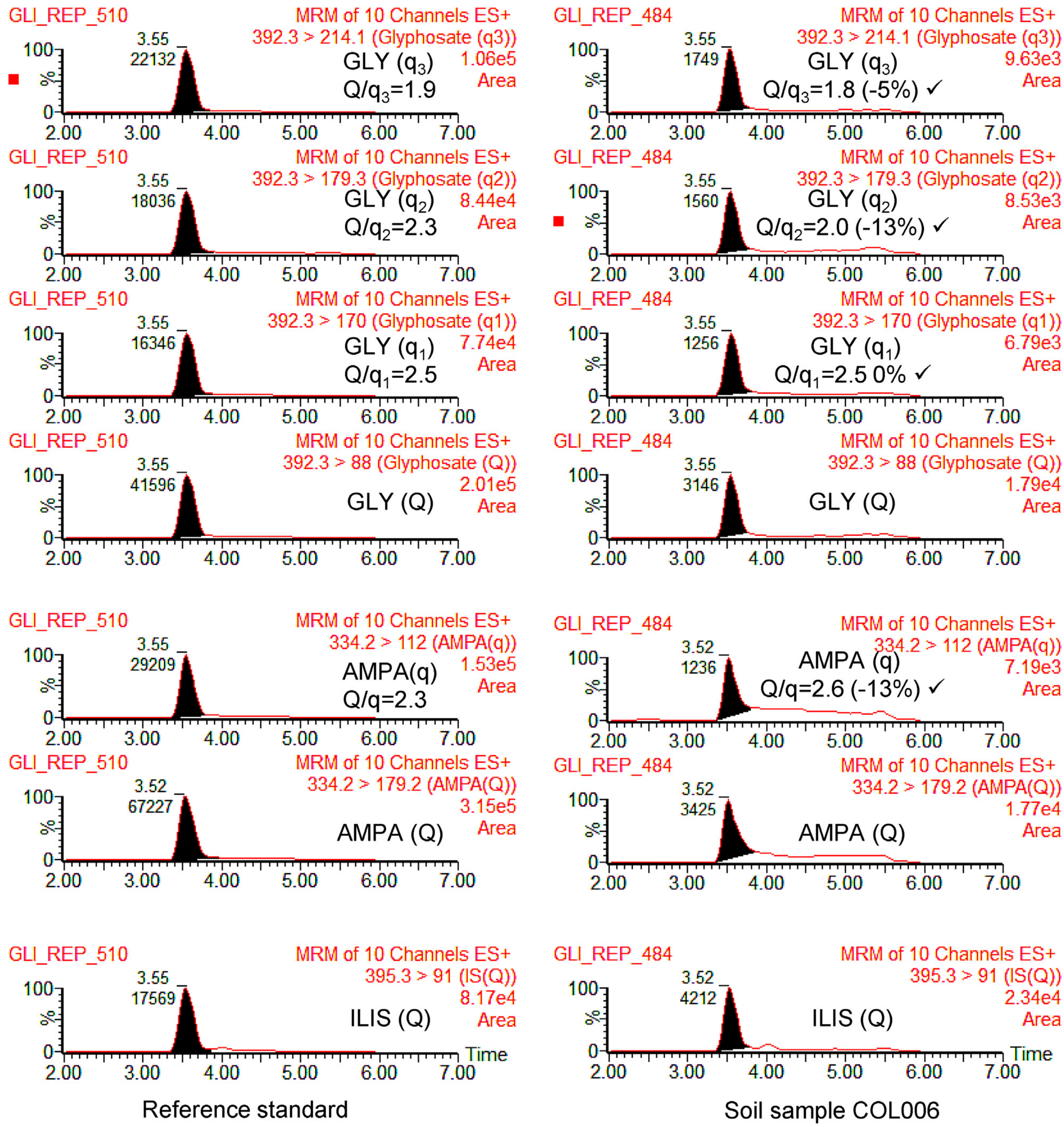

Fig. 4. LC–MS/MS chromatograms of GLY-FMOC, AMPA-FMOC and ILIS-FMOC for reference standards (left) and for the soil sample COL006, containing 1.74 mg/kg of GLY and 1.14 mg/kg of AMPA (right).

Positive findings need to be confirmed by at least two MS/MS transitions and one Q/q ratio, which should fit with that of reference standards with maximum deviations between 10% and 50% depending on the Q/q ratio value [26,28]. The agreement of retention time is also required, with deviations lower than 2.5% respect to a reference standard. In this work, the tolerances for GLY were 25% for Q/q_1 (ion ratio 2.6) and 20% for Q/q_3 (ion ratio 1.9). Q/q_2 was not used for confirmation as it was interfered in several samples. Using Q/q_1 ratio the identity of GLY could be confirmed in the Colombian soils. In two samples the ion ratio deviations were out of tolerance (COL001, COL002), but the identity of GLY was confirmed making use of Q/q_3 ratio (Table 3). The confirmation of GLY in the soils from Argentina was more problematic, as Q/q_1 seemed to be interfered in 6 out of 10 positive samples. Again, the use of an alternative transition (Q/q_3) allowed a reliable confirmation in all the samples. In all cases, the retention time was within ±2.5% deviation. In summary, GLY could be finally confirmed in all soils where it was detected.

The situation for AMPA was more problematic in terms of confirmation, as only two transitions were available, and the quantification transition (334 > 179) seemed to be interfered in most of the soil samples. This led to Q/q ratios higher than expected, above the maximum admissible deviation of ±25% (theoretical Q/q ratio was 2.6). Thus, ion ratio for AMPA was out of tolerance in 7 out of 15 the Colombian soils, and in 2 out of 11 Argentina soils. In these cases, we did not found an alternative criterion for confirmation, although the compliance of retention time, as well as the presence of chromatographic peaks at the two transitions acquired, together with the presence of GLY in these samples, suggested that the compound detected was AMPA actually. More research seems necessary at this point, as well as to accept the possibility to admit higher tolerances in difficult analyte/matrix combinations like occurs in the residue determination of GLY and AMPA in soils. For example, enlarging the tolerance up to 50%, as recently suggested [34], 8 out of the 9 soils that could not be previously confirmed for AMPA would be now confirmed. The fact that the transition used for AMPA quantification (the most sensitive one, 334 > 179) was interfered in some of the soils is particularly problematic, as it does not only affect to confirmation but also to quantification. To evaluate the relevance of this fact, the soils that were out of ion ratio tolerance (marked as "×" in Table 3) were re-quantified using the *q* transition (less sensitive but more selective). In most of these soils, AMPA concentration resulting from *q* calculation was lower than using the *Q* transition, although both data were rather similar. This support our hypothesis on that quantification transition (*Q*) was interfered in these soils leading to positive errors in quantification. Under this situation, we suggest the following approach for AMPA quantification in "difficult" soils: (1) perform analysis of soils using the most sensitive transition (334 > 179); (2) calculate *Q*/*q* ratio and check whether it fits the maximum deviations accepted (in our case 25%); if so, report the concentration value obtained; (3) if ion ratio is out of tolerance, recalculate AMPA concentration using the less sensitive *q* transition (334 > 112). As regards confirmation, we suggest to increase the tolerance in ion ratio deviations to 50% as a realistic approach in this type of analysis. The problematic of confirmation identity by ion ratios accomplishment has been recently discussed when using HRMS, and increasing the tolerance has been already suggested in some particular cases [34–36]. Although technical capabilities of triple quadrupole and Orbitrap or QTOF MS are different, and using ion ratios in SRM transitions or in accurate-mass ions is also different, we consider that increasing tolerances in situations like that reported in this article is needed to have a realistic answer to problematic analyte/matrix combinations.

The use of HR MS is an attractive option for confirmation in some particular/problematic cases on the basis of accurate mass measurements. The potential of HR MS (e.g. Orbitrap or (Q)TOF MS) for quantification has been much less explored in the scientific literature, and its lower sensitivity in comparison to LC–MS/MS with triple quadrupole appears as a limitation for quantification purposes. Therefore, the application of HR MS in this particular case needs to be carefully studied to have detailed information on its realistic potential.

As an example of the method performance, Fig. 4 illustrates the detection and confirmation of the two compounds in the soil COL006, which contained 1.74 mg/kg GLY and 1.14 mg/kg AMPA.

4. Conclusions

This work has shown the problems associated to the residue determination of glyphosate and AMPA in "difficult" soils like those frequently found in South America, commonly with high organic matter content. Even using a powerful, sensitive and selective technique like LC–MS/MS, several problems associated to matrix effects (strong ionization suppression) and to the presence of interfering compounds have been observed in most of the soils tested. A clean-up step was required, resulting from a combination of sample extract dilution, adjusting the soil extract to pH 9 and a SPE with OASIS HLB cartridges, followed by derivatization with FMOC. Satisfactory results were obtained for GLY in validation experiments using isotope-labelled glyphosate as ILIS. Most recoveries were also satisfactory for AMPA, although values near 50% were obtained in some samples. Although not tested in this work, the use of AMPA ILIS might be a good option to improve the results for this analyte.

A careful study was made on the presence of matrix components that shared the transitions of GLY, AMPA and even of ILIS used for matrix effects correction. Particularly problematic were those transitions where m/z 179 was used as product ion. This ion comes from the fragmentation of the derivatized molecules and is related with FMOC. Therefore, it is not specific for GLY and AMPA, and should be avoided. The acquisition of all available transitions instead of the two typically acquired in LC–MS/MS methods is recommended. In any case, confirmation of the identity by ion ratio criterion seems problematic in this type of analysis. The use of wider ion ratio tolerances than those currently set in the EU may be desirable. QTOF MS is an interesting tool to reveal matrix interferences and/or to help in making an appropriate selection of the ions used in analyses. Its high resolution and mass-accuracy makes feasible the discrimination of matrix interferences. While the strong potential of QTOF MS is widely recognized in qualitative analysis, its applications for quantification purposes have been rather limited until now. Despite its lower sensitivity in comparison with LC–MS/MS, LC–QTOF MS appears as an attractive tool to face the analysis of glyphosate in soils, and it is worth to be investigated in the near future.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.chroma.2012.12.007.

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